

DHRUV CHHABRA

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WORK EXPERIENCE

Quantitative Trader - Commodities; Alpha Alternatives Pvt Ltd May 2024 – Sept 2025

- **Multi-Asset Pairs Trading & Risk Framework**

- * Collaborated with discretionary traders to formulate an **early mean-reversion signal** strategy
- * Used gradient-based momentum to capture trend reversal, with **volatility-adjusted stop-loss** to prevent tail-risk
- * Achieved an overall Profit Factor of 1.9 and 45% win rate on portfolio of ₹6 Cr, **generating ₹20 lakh+** in year 2025 alone
- * Took ownership and completed the project after the intern's exit, having reviewed and shaped their initial work

- **Algorithmic Strategy Refinement and Portfolio Optimisation system**

- * Led a **strategic audit** of a trend-following algorithm for commodities' Futures, rectifying critical backtesting biases
- * Automated signal generation framework, **reducing operational errors and slippage cost** from 0.15% to 0.02%
- * Applied Markowitz Mean-Variance theory to rebalance portfolio weights, **boosting monthly returns** to 1.38%

Associate Data Scientist; Optum Global Solutions Ltd

Aug 2021 – May 2024

- **Automated Clinical Documentation & Validation System**

- * Developed a Gen AI engine to automate medical claim appeal summaries, achieving **73% coverage** of all appeal categories
- * Built a correctness evaluation ML model to score AI outputs, **reducing manual audit requirement** by 85%
- * Designed a **double-blind validation tool** to collect data and benchmark AI fidelity against clinician-written summaries
- * Validated **AI non-inferiority** via 2-sample t-test, **reducing documentation time** by 55–60%

- **Prediction of Risk of Hospital Readmission**

- * Worked on a 30-day readmission problem to identify high-risk patients and reduce avoidable hospital costs
- * Built a risk-scoring model on an imbalanced population (**23% incidence**), achieving actionable predictive performance
- * Presented **cost to benefit analysis** for a specific use case to quantify model's utility to senior leadership

- **Prediction of First Occurrence of Multiple Diseases Simultaneously**

- * Designed a neural network using medical history to predict for the very first occurrence of 63 chronic diseases at once
- * Presented the results as proof of concept to the ML and AI R&D team under Optum Clinical Manager

PROJECTS

The Mental Health Coverage Gap in Indian Health Insurance 📄

May 2026

- Quantified gap in India's mental health insurance: even best-in-class retail policies cover close to 0% of outpatient therapy
- Proposed B2B benefits platform redesign with hybrid funding to close the gap that retail insurance economically cannot

Strategic Roadmap: Protecting Margins Amidst Global Silver Shortages 📄

Apr 2026

- Identified structural EBITDA risk for Solar PV manufacturers from silver supply constraints and stress-tested margin impact
- Developed a strategic response with financial hedging, direct offtake agreements with primary miners, and silver intensity R&D, to shift manufacturers from variable to predictable cost structures

Algorithmic Trading with Backtesting Analysis

Jan 2024

- Developed systematic trading strategies (alphas) using technical analysis indicators to generate buy/sell signals
- Back-tested the results against S&P500 as the benchmark and computed the draw-downs and risk-adjusted returns

Option Pricing for an FX Derivative

Dec 2023

- Implemented the Black-Scholes Model to price an FX call option using: the analytical solution, finite difference method, and Monte Carlo simulation for asset prices with discretized time intervals
- Analysed the option Greeks like Delta, Gamma and Vega, and plotted the parabolic option prices for each method

Time Series Analysis of stock price movement

Prof. A Mitra | Dec 2020

- Conducted tests for stationarity and seasonality, and evaluated ACF and PACF plots to select order for the model
- Fitted an ARIMA model on transformed data and conducted residual analysis to verify random distribution of errors

SKILLS

Python, SQL, PySpark, LLM/Gen AI tooling, Git, C/C++, L^AT_EX, Advanced Excel (VBA)

EDUCATION

MSc, Statistics	Indian Institute of Technology Kanpur	8.33 /10	July 2019 - May 2021
BSc, Statistics (Hons.)	PGDAV College, University of Delhi	9.47 /10	July 2016 - May 2019

KEY COURSES

Financial Statistics Statistical Methods for Business Analytics Statistical Simulation & Inference Econometrics

EXTRA CURRICULAR

- Awarded the **Junior Diploma** in Hindustani Classical Music (**Violin**) from Prayag Sangeet Samiti, 2025
- Qualified the **Goethe-Zertificat A2** level **German** language examination from Goethe-Institut Delhi, 2024
- Awarded **1st Best Speaker** in **Novice Debating League** organised by DebSoc IIT Kanpur, 2020